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Optimizing Barter Trading Networks
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Bartering, the oldest form of trade, still thrives today. An estimated $14B worth of goods is exchanged annually in barter networks. This is despite the networks not solving the age-old Barter Trading Matching Problem (BTMP): find a trader that has what I want and wants what I have.

I developed five algorithms that find the single trader (denoted as BTMP1) or the set of traders (BTMPSet) that can exchange the maximum amount of items given a trader T.
The algorithms solve BTMP in different ways. One uses network flow, two use inverted-indices, one uses databases, and the last is brute-force.

I tested the algorithms on 16 datasets modeled after today's networks (e.g., Craigslist, Rehash). Our inverted-index based BTMP1 algorithm returns optimal trade proposals within 2ms on average that exchange 8% of a trader's inventory. For BTMPSet, our network-flow based algorithm returns optimal trade proposals within 13ms that exchange 83% of a trader's inventory. The heuristic inverted-index based BTMPSet algorithm returns proposals that exchange 73% of inventory but using 35% less traders.

I show through complexity analysis that the network flow and the inverted-index based algorithms scale with the ratio N/M (instead of N or M), where N is the number of traders, and M is the number of items in the network. And through experimentation, I show that these algorithms can handle sizes beyond the sizes of today's networks.

Using network flow and inverted-indices, I created scalable algorithms that make finding possible traders in barter networks much easier.